Optimization and Curve Sketching

Section 1: Critical Points and First Derivative Test

1 Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. $x = 1$ and $x = 2$

b. $x = -1$ and $x = -2$

c. $x = 3$ and $x = 0$

d. $x = 2$ only

2 Classify the local extrema of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. Local max at $x = 1$; local min at $x = 2$

b. Local min at $x = 1$; local max at $x = 2$

c. Inflection point at $x = 1$

d. Local max at $x = 2$ only

3 Find the absolute maximum of $g(x) = 4x - x^2$ on the closed interval $[0, 3]$.

a. 4

b. 3

c. 0

d. 5

Section 2: Concavity and the Second Derivative Test

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4 Determine the interval(s) where $f(x) = x^4 - 4x^3 + 10$ is concave upward.

a. $(-\infty, 0) \cup (2, \infty)$

b. $(0, 2)$

c. $(2, \infty)$

d. $(-\infty, 2)$

5 Find the x-coordinates of all inflection points of $y = x^3 - 6x^2 + 9x$.

a. $x = 2$

b. $x = 1$ and $x = 3$

c. $x = 0$

d. $x = 4$

6 If $f'(3) = 0$ and $f''(3) = -5$, what does the Second Derivative Test conclude about $x = 3$?

a. f has a local maximum at $x = 3$

b. f has a local minimum at $x = 3$

c. $x = 3$ is an inflection point

d. The test is inconclusive

Section 3: Applied Optimization: Perimeter, Area, and Volume

7 A farmer has 120 m of fencing to enclose a rectangular field adjacent to a river (no fence needed along the river). What dimensions maximize the enclosed area?

a. Width $= 30$ m, Length $= 60$ m

b. Width $= 40$ m, Length $= 40$ m

c. Width $= 20$ m, Length $= 80$ m

d. Width $= 25$ m, Length $= 70$ m

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8 An open-top rectangular box is made from a 16 by 10 inch cardboard sheet by cutting equal squares of side x from each corner and folding up the sides. What x maximizes the box volume?

a. 2 inches

b. 3 inches

c. 1.5 inches

d. 2.5 inches

9 A closed cylindrical can must hold $1000\ \text{cm}^3$ of liquid. What relationship between height h and radius r minimizes total surface area?

a. $h = 2r$

b. $h = r$

c. $h = 4r$

d. $h = \pi r$

Section 4: Economics and Distance Optimization

10 A company's revenue and cost functions are $R(x) = 50x - 0.5x^2$ and $C(x) = 10x + 200$. What production level x maximizes profit?

a. 40

b. 50

c. 30

d. 60

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11 Find the point on the line $y = 2x + 1$ that is closest to the point $(4, 4)$.

a. $(2, 5)$

b. $(1, 3)$

c. $(3, 7)$

d. $(1.5, 4)$

12 A Norman window has the shape of a rectangle surmounted by a semicircle. If the perimeter is 16 ft, what radius r of the semicircle maximizes the light admitted?

a. $\frac{16}{4+\pi}$ ft

b. $\frac{8}{2+\pi}$ ft

c. 2 ft

d. $\frac{16}{\pi}$ ft

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